

DENSE CHAINS, ANTICHAINS, AND UNIVERSAL PARTIAL ORDERS INSIDE A BOUNDED FINITE-ONE DEGREE

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ABSTRACT. We construct a nonrecursive set $A \leq_T \emptyset'$ and a uniformly computable family of sets $C_0, C_1, \dots$, all bounded finite-one equivalent to A , such that the corresponding 1-degrees form a copy of the dense linear order $(\mathbb{Q}, \leq)$. Motivated by a recent preprint of Richter, Stephan, and Zhang, which shows that bounded finite-one degrees can be as rigid as a discrete ω -chain and asks whether there are bounded finite-one degrees consisting exactly of a dense linearly ordered set of 1-degrees, we introduce a block-density profile method for controlling one-one reducibility inside a single bounded finite-one degree.

As further applications, in the same bounded finite-one degree we obtain an infinite antichain of 1-degrees and, more generally, an embedded copy of every countable partial order. A single bounded finite-one degree can already exhibit dense, incomparable, and universal order-theoretic behaviour.

Our main technical tool is a profile theorem based on computable block-density codings. The witness set constructed here is not m -rigid, so the phenomena obtained in this paper arise from a mechanism different from earlier m -rigidity-based constructions. Although our results do not settle the exact realization problem posed by Richter, Stephan, and Zhang, we show that density itself is not the obstruction: a single bounded finite-one degree may already contain a copy of $(\mathbb{Q}, \leq)$, an infinite antichain, and embeddings of all countable partial orders.

Personal Website Version. This version is mathematically identical to the corresponding arXiv version. No changes have been made to the statements, proofs, definitions, or mathematical arguments. The only modifications concern the acknowledgments section, where the role of the AI systems in the development of the work is described in greater detail.

1. INTRODUCTION

Strong refinements of many-one reducibility, such as one-one, finite-one, and bounded finite-one reducibility, provide a natural framework for analysing the internal structure of many-one degrees. Given a set A , one may ask which partially ordered configurations of 1-degrees can occur inside a fixed finite-one degree or bounded finite-one degree represented by A .

This theme was investigated in detail by Richter, Stephan, and Zhang [13]. On the finite-one side, they showed that nonrecursive, nonirreducible finite-one degrees are already highly complex: every computable partial order embeds effectively into the structure of 1-degrees inside such a finite-one degree, and hence every countable partial order embeds noneffectively. By contrast, their bounded finite-one results reveal a much more delicate picture. In particular, they constructed a nonrecursive bounded finite-one degree whose internal 1-degrees form a strictly ascending chain of order type ω [13, Theorem 18]. The finite-one universality phenomenon therefore does not extend automatically to the bounded finite-one setting.

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This led Richter, Stephan, and Zhang to ask which linear orders can occur *exactly* as the full structure of 1-degrees inside a bounded finite-one degree. Their Open Problem 3 [13, Section 5] is:

Are there bounded finite-one degrees consisting exactly of a dense linearly ordered set of 1-degrees?

In their further remarks, Richter, Stephan, and Zhang go beyond Open Problem 3 and ask more generally whether bounded finite-one degrees can exhibit linear orders other than the order type ω produced by their Theorem 18. In the strong exact-realization sense made explicit there, our results do not answer this question affirmatively, since the bounded finite-one degree constructed here is not itself exactly linearly ordered. We show, however, that the bounded finite-one setting already supports much richer linear behaviour internally: a single bounded finite-one degree may contain a copy of $(\mathbb{Q}, \leq)$, and indeed, by Corollary 3.11, embedded copies of all countable linear orders.

A different perspective on bounded finite-one degrees was developed in recent work on m -rigidity [3]. Recall that a set A is m -rigid if every total computable many-one self-reduction of A is eventually the identity. In that setting, one obtains strong negative information about typical bounded finite-one degrees: for a measure-1 and comeager class of sets, a single bounded finite-one degree already contains an infinite antichain of 1-degrees, and therefore cannot be linearly ordered [3]. In the typical rigid regime, Open Problem 3 therefore has a negative answer.

The present paper moves in the opposite direction. Rather than studying the typical rigid case, we construct explicitly a nonrecursive set $A \leq_T \emptyset'$ whose bounded finite-one degree exhibits a rich internal order structure. Our main technical tool is Theorem 3.2, a profile theorem based on computable block-density codings. Roughly speaking, each representative is assigned a computable block-density profile recording its local preimage density over a common block decomposition. Pointwise domination of profiles yields positive 1-reductions, while a uniform positive gap along infinitely many selected blocks allows us to diagonalize directly against 1-reductions in the opposite direction. This produces a flexible block-geometric mechanism for prescribing comparison patterns while keeping all representatives inside a single bounded finite-one degree.

As consequences, we show that one can realize within a single bounded finite-one degree a copy of the dense linear order $(\mathbb{Q}, \leq)$, an infinite antichain of 1-degrees, and, more generally, an embedded copy of every countable partial order; see Corollaries 3.4, 3.5, 3.6, 3.10, and 3.11. In particular, although we do not solve the exact realization problem posed by Richter, Stephan, and Zhang, we show that density itself is not the obstruction: a bounded finite-one degree may already contain a copy of $(\mathbb{Q}, \leq)$, and indeed may support universal countable order-theoretic behaviour.

It is important that the witness constructed here lies outside the rigidity framework of [3]. The set A obtained in Corollary 3.6 is constant on a computable partition into finite blocks and therefore admits nontrivial computable many-one self-reductions collapsing each block to a distinguished representative; see Remark 3.9. In particular, A is not m -rigid. Since the class of m -rigid sets has measure 1 and is comeager [3], our witness necessarily belongs to the null and meager complement of the typical rigid regime. The antichain structure arising here is therefore not a consequence of m -rigidity. Rather, it coexists with dense linear behaviour inside a single bounded finite-one degree.

Accordingly, the contribution of this paper is not an affirmative solution to Open Problem 3 in its exact form. The degree we construct does not consist *exactly* of a dense linear order of 1-degrees, since it also contains incomparable elements. What our results show is that the true difficulty of the open problem lies in the word “exactly”: bounded finite-one degrees may already contain dense chains, infinite antichains, and even universal countable partial orders, all inside a single explicitly constructed degree.

The paper is organised as follows. After fixing notation, we prove the Profile Theorem, which gives a general block-density method for producing prescribed one-one comparison patterns inside a single bounded finite-one degree. We then derive results yielding a copy of $(\mathbb{Q}, \leq)$, an infinite antichain, their coexistence within the same degree, and finally universal embeddings of computable and countable partial orders.

2. PRELIMINARIES AND NOTATION

We assume familiarity with standard notions and notation from computability theory; for general background see [4, 7, 8, 9, 12, 14]. For background more specifically on strong reducibilities and degree structures, see also [10, 11]. All sets considered are subsets of the natural numbers $\mathbb{N} = \{0, 1, \dots\}$. We routinely identify a set $A \subseteq \mathbb{N}$ with its characteristic function, writing $A(x) = 1$ if $x \in A$ and $A(x) = 0$ otherwise.

We fix a standard effective enumeration $\varphi_0, \varphi_1, \dots$ of all partial computable functions. We write $\varphi_e(x) \downarrow$ to indicate that the e -th partial computable function converges on input x . We denote by $\emptyset'$ the Turing jump of the empty set. Any set constructed by a stage construction using finitely many queries to $\emptyset'$ at each stage is computable in $\emptyset'$, and hence belongs to Δ_2^0 .

We now introduce the reducibilities considered in this paper, which are variants of many-one reducibility. Recall that a set $A \subseteq \mathbb{N}$ is many-one reducible to a set $B \subseteq \mathbb{N}$ if there is a total computable function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$x \in A \iff f(x) \in B \quad \text{for all } x \in \mathbb{N}.$$

Definition 2.1. Let $A, B \subseteq \mathbb{N}$.

- (i) We write $A \leq_1 B$ if there is a total computable injective function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$x \in A \iff f(x) \in B \quad \text{for all } x \in \mathbb{N}.$$

- (ii) We write $A \leq_{\text{bfo}} B$ if there are a total computable function $f : \mathbb{N} \rightarrow \mathbb{N}$ and a constant $c \geq 1$ such that

$$x \in A \iff f(x) \in B \quad \text{for all } x \in \mathbb{N},$$

and

$$|f^{-1}(y)| \leq c \quad \text{for all } y \in \mathbb{N}.$$

We say that A and B are r -equivalent, written $A \equiv_r B$, if both $A \leq_r B$ and $B \leq_r A$, where $r \in \{1, \text{bfo}\}$. The equivalence classes are the r -degrees. We write $A <_r B$ if $A \leq_r B$ but $B \not\leq_r A$.

By definition,

$$A \leq_1 B \implies A \leq_{\text{bfo}} B.$$

Thus a bounded finite-one degree generally contains a nontrivial partially ordered structure of one-one degrees.

3. A PROFILE THEOREM FOR BLOCK-DENSITY CODINGS

This section isolates the combinatorial core of our method. The basic idea is to encode each representative by a computable block-density profile measuring its local preimage density over a common factorial block decomposition. Pointwise comparison of profiles yields positive 1-reductions, while a uniform positive gap along infinitely many selected blocks provides enough room for direct diagonalization against 1-reductions in the opposite direction. The following Profile Theorem packages this mechanism in a general form from which all later applications will be derived.

Definition 3.1. A *block-density profile* is a total function

$$u : \mathbb{N} \rightarrow \mathbb{Q} \cap (1, 2).$$

A family $\{u_i\}_{i \in \mathbb{N}}$ of block-density profiles is *uniformly computable* if the function

$$(i, k) \mapsto u_i(k)$$

is computable.

Informally, a block-density profile records, block by block, the local preimage density used in the coding construction.

Theorem 3.2 (Profile Theorem). *Let $\{u_i\}_{i \in \mathbb{N}}$ be a uniformly computable family of block-density profiles, and let $D \subseteq \mathbb{N}^2$ be a computable set of ordered pairs. Assume that for each $(i, j) \in D$ we are given, uniformly computably,*

- (i) *a rational number $\delta_{i,j} > 0$; and*
- (ii) *a strictly increasing computable function $s_{i,j} : \mathbb{N} \rightarrow \mathbb{N}$ with infinite range $S_{i,j} = \{s_{i,j}(t) : t \in \mathbb{N}\}$,*

such that

$$u_i(k) \geq u_j(k) + \delta_{i,j} \tag{1}$$

for every $k \in S_{i,j}$.

Then there exist a set $A \subseteq \mathbb{N}$, computable in $\mathcal{O}$, and a uniformly computable family of total surjective functions

$$h_i : \mathbb{N} \rightarrow \mathbb{N} \quad (i \in \mathbb{N}),$$

such that, writing $I_k = [a_k, a_{k+1})$ with $a_0 = 0$ and $a_{k+1} = a_k + k!$, the following hold:

- (a) *A is constant on each block I_k .*
- (b) *Each h_i is bounded finite-one with global bound 2.*
- (c) *For $C_i(x) = A(h_i(x))$, one has $C_i \equiv_{\text{bfo}} A$ for every $i \in \mathbb{N}$.*
- (d) *If $u_i(k) \leq u_j(k)$ for every $k \in \mathbb{N}$, then $C_i \leq_1 C_j$.*
- (e) *If $(i, j) \in D$, then $C_i \not\leq_1 C_j$.*

In particular, whenever $u_i(k) \leq u_j(k)$ for all $k \in \mathbb{N}$ and $(j, i) \in D$, one has $C_i <_1 C_j$.

Proof. For each $k \in \mathbb{N}$, let

$$a_0 = 0, \quad a_{k+1} = a_k + k!, \quad I_k = [a_k, a_{k+1}).$$

Thus $\mathbb{N}$ is partitioned into contiguous blocks $\{I_k\}_{k \in \mathbb{N}}$ with $|I_k| = k!$. We construct $A \subseteq \mathbb{N}$ so that it is constant on each I_k .

For each $i, k \in \mathbb{N}$, define

$$\ell_i(k) = \lfloor u_i(k) k! \rfloor.$$

Since $1 < u_i(k) < 2$, we have

$$k! \leq \ell_i(k) < 2k! \quad \text{for all } i, k.$$

Now let

$$b_0^i = 0, \quad b_{k+1}^i = b_k^i + \ell_i(k), \quad K_{i,k} = [b_k^i, b_{k+1}^i).$$

Thus for each fixed i , the family $\{K_{i,k}\}_{k \in \mathbb{N}}$ is a computable partition of $\mathbb{N}$, and

$$|K_{i,k}| = \ell_i(k).$$

Define a total computable function $h_i : \mathbb{N} \rightarrow \mathbb{N}$ blockwise as follows: if $x = b_k^i + r$ with $0 \leq r < \ell_i(k)$, set

$$h_i(x) = a_k + \left\lfloor \frac{r k!}{\ell_i(k)} \right\rfloor.$$

For fixed i, k , consider the function

$$g_{i,k}(r) = \left\lfloor \frac{r k!}{\ell_i(k)} \right\rfloor \quad (0 \leq r < \ell_i(k)).$$

Since $k! \leq \ell_i(k)$, we have $k!/\ell_i(k) \leq 1$, so $g_{i,k}$ is nondecreasing and

$$g_{i,k}(r+1) - g_{i,k}(r) \leq 1 \quad \text{for all } r < \ell_i(k) - 1. \quad (2)$$

Also,

$$g_{i,k}(0) = 0$$

and

$$g_{i,k}(\ell_i(k) - 1) = \left\lfloor \frac{(\ell_i(k) - 1)k!}{\ell_i(k)} \right\rfloor = \left\lfloor k! - \frac{k!}{\ell_i(k)} \right\rfloor = k! - 1.$$

Since $g_{i,k}$ is nondecreasing, integer-valued, and changes by at most 1 at each step by (2), its range is exactly $\{0, 1, \dots, k! - 1\}$. Since

$$h_i(b_k^i + r) = a_k + g_{i,k}(r) \quad (0 \leq r < \ell_i(k)),$$

it follows that

$$h_i[K_{i,k}] = \{a_k + q : 0 \leq q < k!\} = I_k.$$

Since $h_i[K_{i,k}] = I_k$ for every k and the blocks I_k partition $\mathbb{N}$, the function h_i is surjective.

Moreover, for each $q < k!$, the fiber $g_{i,k}^{-1}(q)$ consists of those r such that

$$q \leq \frac{r k!}{\ell_i(k)} < q + 1,$$

that is,

$$\frac{q \ell_i(k)}{k!} \leq r < \frac{(q+1)\ell_i(k)}{k!}.$$

This interval has length

$$\frac{\ell_i(k)}{k!} < 2,$$

so it contains at most two integers.

Thus every fiber of the restriction

$$h_i \upharpoonright K_{i,k} : K_{i,k} \rightarrow I_k$$

has size at most 2. Now let $y \in \mathbb{N}$, and let k be the unique index such that $y \in I_k$. Since $h_i(K_{i,m}) \subseteq I_m$ for every m , and the blocks I_m are pairwise disjoint, any preimage of y under h_i must lie in $K_{i,k}$. Hence

$$|h_i^{-1}(y)| \leq 2.$$

Hence each h_i is bounded finite-one with global bound 2, proving item (b).

Finally, since $(i, k) \mapsto \ell_i(k)$ is computable, the endpoints $(i, k) \mapsto b_k^i$ are uniformly computable. Therefore, given (i, x) , we can effectively find the unique k such that $x \in K_{i,k}$ and then compute $h_i(x)$. So the family $\{h_i\}_{i \in \mathbb{N}}$ is uniformly computable.

Once A has been constructed, define

$$C_i(x) = A(h_i(x)).$$

We first verify item (c), which holds for every set A , and item (d), which holds for any set A that is constant on each block I_k .

Since h_i is total computable and bounded finite-one,

$$C_i \leq_{\text{bfo}} A.$$

Conversely, because h_i is computable and surjective, the function

$$r_i(y) = \mu x [h_i(x) = y]$$

is total computable. It is injective, since two distinct values of y cannot have the same least preimage. Therefore

$$C_i(r_i(y)) = A(h_i(r_i(y))) = A(y),$$

so $A \leq_1 C_i$. Hence

$$C_i \equiv_{\text{bfo}} A \quad \text{for every } i \in \mathbb{N}.$$

Thus item (c) holds for every set A .

Now suppose that $u_i(k) \leq u_j(k)$ for every k . Then $\ell_i(k) \leq \ell_j(k)$ for every k . Define a total computable function $f_{i,j} : \mathbb{N} \rightarrow \mathbb{N}$ blockwise by

$$f_{i,j}(b_k^i + r) = b_k^j + r \quad (0 \leq r < \ell_i(k)).$$

Since the partition $\{K_{i,k}\}_{k \in \mathbb{N}}$ is computable, given x we can find the unique k and r such that

$$x = b_k^i + r, \quad 0 \leq r < \ell_i(k).$$

Because $\ell_i(k) \leq \ell_j(k)$, the value

$$f_{i,j}(x) = b_k^j + r$$

is well-defined. On each block $K_{i,k}$ the function $f_{i,j}$ is a translation, and distinct blocks are mapped into distinct blocks, so $f_{i,j}$ is injective.

If $x \in K_{i,k}$, then $f_{i,j}(x) \in K_{j,k}$, so

$$h_i(x) \in I_k \quad \text{and} \quad h_j(f_{i,j}(x)) \in I_k.$$

Since A is constant on I_k and both $h_i(x)$ and $h_j(f_{i,j}(x))$ belong to I_k , we have

$$C_i(x) = A(h_i(x)) = A(h_j(f_{i,j}(x))) = C_j(f_{i,j}(x)).$$

Thus

$$C_i \leq_1 C_j.$$

This proves that item (d) holds for any A that is constant on each block I_k .

It remains to construct A so as to satisfy (a) and (e). For every $e \in \mathbb{N}$ and every $(i, j) \in D$, let

$$R_{e,i,j} : \varphi_e \text{ is not a total one-one reduction from } C_i \text{ to } C_j.$$

Fix an effective enumeration of all such requirements.

We build A in stages using $\emptyset'$. Let M_s denote the largest block index already defined before stage s , with $M_0 = -1$. Thus, before stage s , every block I_m with $m \leq M_s$ has already been assigned a value. At the end of stage s , every block I_m with $m \leq M_{s+1}$ will have been assigned a value, and later stages will never change these values.

Suppose the s -th requirement is $R_{e,i,j}$. We search for a number $t \in \mathbb{N}$ such that, writing

$$k = s_{i,j}(t),$$

we have

$$k > M_s \text{ and } |K_{i,k}| > \sum_{m=0}^k |K_{j,m}|. \quad (3)$$

This search terminates. Indeed, for every $k \in S_{i,j}$, by (1),

$$|K_{i,k}| - |K_{j,k}| = \lfloor u_i(k)k! \rfloor - \lfloor u_j(k)k! \rfloor \geq \delta_{i,j}k! - 1. \quad (4)$$

On the other hand, for every m ,

$$|K_{j,m}| < 2m!,$$

so

$$\sum_{m=0}^{k-1} |K_{j,m}| < 2 \sum_{m=0}^{k-1} m!.$$

Since for $k \geq 1$,

$$\sum_{m=0}^{k-1} m! \leq (k-1)! \sum_{r=0}^{k-1} \frac{1}{r!} < 3(k-1)!,$$

we obtain

$$\sum_{m=0}^{k-1} |K_{j,m}| < 6(k-1)!. \quad (5)$$

Because

$$\delta_{i,j}k! = \delta_{i,j}k(k-1)!,$$

the quantity $\delta_{i,j}k! - 1$ eventually dominates $6(k-1)!$, so by (4) and (5) we obtain

$$|K_{i,k}| - |K_{j,k}| > \sum_{m < k} |K_{j,m}|$$

which implies

$$|K_{i,k}| > |K_{j,k}| + \sum_{m < k} |K_{j,m}| = \sum_{m \leq k} |K_{j,m}|.$$

Hence, for all sufficiently large $k \in S_{i,j}$, we have

$$|K_{i,k}| > \sum_{m \leq k} |K_{j,m}|,$$

and also $k > M_s$. Equation (3) therefore holds for all sufficiently large $k \in S_{i,j}$.

Since $S_{i,j}$ is infinite and computable, the search for t terminates.

Having chosen such a k , we act as follows.

- [Step 1] Use $\emptyset'$ to decide whether $\varphi_e(x) \downarrow$ for every $x \in K_{i,k}$. If not, then φ_e is not total, so $R_{e,i,j}$ is already satisfied. In this case, for every previously undefined block I_m with $M_s < m \leq k$, define

$$A(z) = 0 \quad \text{for all } z \in I_m,$$

and set $M_{s+1} = k$.

- [Step 2] Suppose now that $\varphi_e(x) \downarrow$ for every $x \in K_{i,k}$. Since $K_{i,k}$ is finite, we can compute all values $\varphi_e(x)$ for $x \in K_{i,k}$. If these values are not pairwise distinct, then φ_e is not injective, so $R_{e,i,j}$ is satisfied. In that case, for every previously undefined block I_m with $M_s < m \leq k$, define

$$A(z) = 0 \quad \text{for all } z \in I_m,$$

and set $M_{s+1} = k$.

- [Step 3] Suppose finally that φ_e is total and injective on $K_{i,k}$. By (3),

$$|K_{i,k}| > \left| \bigcup_{m \leq k} K_{j,m} \right|.$$

Since φ_e is injective on $K_{i,k}$,

$$|\varphi_e[K_{i,k}]| = |K_{i,k}|,$$

so $\varphi_e[K_{i,k}]$ cannot be contained in $\bigcup_{m \leq k} K_{j,m}$.

Hence there exists some $x \in K_{i,k}$ such that

$$y = \varphi_e(x) \in K_{j,\ell}$$

for some $\ell > k$. Fix the least such x , and let ℓ be the corresponding index. Define

$$A(z) = 1 \quad \text{for all } z \in I_k, \quad A(z) = 0 \quad \text{for all } z \in I_\ell,$$

and for every previously undefined block I_m with $M_s < m < \ell$ and $m \neq k$, define

$$A(z) = 0 \quad \text{for all } z \in I_m.$$

Finally let $M_{s+1} = \ell$.

We now verify the construction by checking that every requirement $R_{e,i,j}$ with $(i,j) \in D$ is satisfied. Fix $e \in \mathbb{N}$ and $(i,j) \in D$. Let s be the stage at which the requirement $R_{e,i,j}$ is considered. In Steps 1 and 2, the requirement $R_{e,i,j}$ is satisfied because φ_e is not total in the first case and not injective in the second. In Step 3, with $x \in K_{i,k}$ and $y = \varphi_e(x) \in K_{j,\ell}$, we have

$$h_i(x) \in I_k \quad \text{and} \quad h_j(y) \in I_\ell.$$

Since A is constant with value 1 on I_k and constant with value 0 on I_ℓ , we obtain

$$C_i(x) = A(h_i(x)) = 1 \quad \text{and} \quad C_j(\varphi_e(x)) = C_j(y) = A(h_j(y)) = 0.$$

Hence

$$C_i(x) \neq C_j(\varphi_e(x)),$$

so φ_e is not a one-one reduction from C_i to C_j . Since e and $(i,j) \in D$ were arbitrary, every requirement $R_{e,i,j}$ is satisfied. Therefore item (e) holds.

At every stage, values are assigned only to blocks I_m with $m > M_s$, so there is no injury. Moreover, the sequence $(M_s)_{s \in \mathbb{N}}$ is strictly increasing, and at the end of stage s all blocks I_m with $m \leq M_{s+1}$ have been defined. Hence every block I_m is defined at some finite stage. Thus A is total and constant on each block, which gives item (a).

To compute $A(n)$ from $\emptyset'$, first compute the unique m such that $n \in I_m$. Then, using $\emptyset'$, simulate the construction stage by stage until the end of some stage s with $M_{s+1} \geq m$. By the no-injury property, the value assigned to the block I_m has then stabilized permanently, so we can output $A(n)$. Therefore $A \leq_T \emptyset'$. This completes the proof. $\square$

Remark 3.3. In the situation of Theorem 3.2, the 1-degree of A is a common lower bound of the family $\{\deg_1(C_i)\}_{i \in \mathbb{N}}$. Indeed, for every i , the least-preimage function

$$r_i(y) = \mu x [h_i(x) = y]$$

is total computable and injective, and satisfies

$$A(y) = C_i(r_i(y)).$$

Hence

$$A \leq_1 C_i \quad \text{for every } i \in \mathbb{N}.$$

The next two corollaries illustrate the two basic regimes of the profile method. Constant profiles yield monotone comparison patterns and hence a dense rational chain, whereas oscillating profiles create local dominance in opposite directions and therefore force pairwise incomparability.

Corollary 3.4 (Constant profiles). *Fix an injective computable enumeration*

$$q_0, q_1, \dots$$

of $\mathbb{Q} \cap (1, 2)$, and define

$$u_i(k) = q_i \quad \text{for all } i, k \in \mathbb{N}.$$

Let

$$D = \{(i, j) \in \mathbb{N}^2 : q_i > q_j\}.$$

For $(i, j) \in D$, define

$$\delta_{i,j} = q_i - q_j \quad \text{and} \quad s_{i,j}(t) = t.$$

Then Theorem 3.2 yields a set $A \leq_T \emptyset'$ and sets

$$B_{q_i} = C_i$$

such that

$$B_{q_i} \equiv_{\text{bfo}} A \quad \text{for every } i \in \mathbb{N},$$

and

$$B_{q_i} <_1 B_{q_j} \quad \text{if and only if } q_i < q_j.$$

Hence the 1-degrees inside the bounded finite-one degree of A contain a copy of $(\mathbb{Q}, \leq)$.

Proof. The hypotheses of Theorem 3.2 are satisfied: the family $u_i(k) = q_i$ is uniformly computable, the set

$$D = \{(i, j) \in \mathbb{N}^2 : q_i > q_j\}$$

is computable, for each $(i, j) \in D$ the rational

$$\delta_{i,j} = q_i - q_j > 0$$

is uniformly computable, and

$$s_{i,j}(t) = t$$

is strictly increasing, computable, and has infinite range. Moreover, for every $(i, j) \in D$ and every k ,

$$u_i(k) = q_i = q_j + (q_i - q_j) = u_j(k) + \delta_{i,j}.$$

It follows from Theorem 3.2 that there exist a set $A \leq_T \emptyset'$ and sets C_i such that $C_i \equiv_{\text{bfo}} A$ for every i . Define

$$B_{q_i} = C_i.$$

If $q_i < q_j$, then $u_i(k) \leq u_j(k)$ for every k , so

$$B_{q_i} \leq_1 B_{q_j}$$

by item (d) of Theorem 3.2. Since $q_j > q_i$, we have $(j, i) \in D$, and therefore

$$B_{q_j} \not\leq_1 B_{q_i}$$

by item (e). Hence

$$B_{q_i} <_1 B_{q_j}.$$

Conversely, suppose that

$$B_{q_i} <_1 B_{q_j}.$$

Then $B_{q_i} \leq_1 B_{q_j}$. If $q_i > q_j$, then $(i, j) \in D$, and item (e) would give

$$B_{q_i} \not\leq_1 B_{q_j},$$

a contradiction. If $q_i = q_j$, then since the enumeration $(q_i)_{i \in \mathbb{N}}$ is injective, we must have $i = j$, hence $B_{q_i} = B_{q_j}$, contradicting $B_{q_i} <_1 B_{q_j}$. Therefore $q_i < q_j$.

Thus

$$B_{q_i} <_1 B_{q_j} \quad \text{if and only if } q_i < q_j.$$

Moreover,

$$B_{q_i} \leq_1 B_{q_j} \iff q_i \leq q_j.$$

Indeed, the implication “ $\Rightarrow$ ” follows from item (e), since $q_i > q_j$ would imply $(i, j) \in D$, while “ $\Leftarrow$ ” follows from item (d) when $q_i < q_j$, and is trivial when $q_i = q_j$. Therefore the function

$$q \mapsto \deg_1(B_q)$$

is an order-embedding of $(\mathbb{Q} \cap (1, 2), \leq)$ into the poset of 1-degrees inside the bounded finite-one degree of A .

Since each $B_{q_i} \equiv_{\text{bfo}} A$, the corresponding 1-degrees all lie inside the bounded finite-one degree of A . The corresponding 1-degrees therefore form a copy of $(\mathbb{Q} \cap (1, 2), \leq)$, and hence a copy of $(\mathbb{Q}, \leq)$. □

The previous corollary used constant profiles to obtain a dense chain. We now turn to the complementary phenomenon: by letting the profiles oscillate between two rational densities on pairwise disjoint computable supports, we force mutual incomparability and thereby obtain an infinite antichain of 1-degrees.

Corollary 3.5 (Oscillating profiles). *Fix rationals $1 < \alpha < \beta < 2$, and for each $n \in \mathbb{N}$ let*

$$Z_n = \{2^n(2t + 1) : t \in \mathbb{N}\}.$$

Then the sets Z_n are pairwise disjoint, infinite, and uniformly computable. Define, for each $n, k \in \mathbb{N}$,

$$u_n(k) = \begin{cases} \beta & \text{if } k \in Z_n, \\ \alpha & \text{if } k \notin Z_n. \end{cases}$$

Let

$$D = \{(n, m) \in \mathbb{N}^2 : n \neq m\}.$$

For $n \neq m$, define

$$\delta_{n,m} = \beta - \alpha \quad \text{and} \quad s_{n,m}(t) = 2^n(2t + 1).$$

Then Theorem 3.2 yields a set $A \leq_T \emptyset'$ and sets

$$W_n = C_n$$

such that

$$W_n \equiv_{\text{bfo}} A \quad \text{for every } n \in \mathbb{N},$$

and

$$W_n \not\leq_1 W_m \quad \text{whenever } n \neq m.$$

Hence $\{\deg_1(W_n)\}_{n \in \mathbb{N}}$ is an infinite antichain of 1-degrees inside a single bounded finite-one degree.

Proof. For each $n \in \mathbb{N}$,

$$Z_n = \{2^n(2t + 1) : t \in \mathbb{N}\}$$

is infinite, and the sets Z_n are pairwise disjoint. The family $\{Z_n\}_{n \in \mathbb{N}}$ is uniformly computable, since

$$k \in Z_n \iff k > 0, \ 2^n \mid k, \text{ and } 2^{n+1} \nmid k.$$

It follows that the family of profiles $\{u_n\}_{n \in \mathbb{N}}$ is uniformly computable. Moreover,

$$D = \{(n, m) \in \mathbb{N}^2 : n \neq m\}$$

is computable. For each $(n, m) \in D$, the rational

$$\delta_{n,m} = \beta - \alpha > 0$$

is uniformly computable, and

$$s_{n,m}(t) = 2^n(2t + 1)$$

is a strictly increasing computable function with infinite range

$$S_{n,m} = Z_n.$$

Now fix $(n, m) \in D$, and let $k \in S_{n,m} = Z_n$. Since $n \neq m$ and the sets Z_n are pairwise disjoint, we have $k \notin Z_m$. Hence

$$u_n(k) = \beta \quad \text{and} \quad u_m(k) = \alpha.$$

Therefore

$$u_n(k) = \beta = \alpha + (\beta - \alpha) = u_m(k) + \delta_{n,m},$$

so condition (1) holds.

Applying Theorem 3.2 and writing

$$W_n := C_n \quad (n \in \mathbb{N}),$$

we obtain a set $A \leq_T \emptyset'$ and sets W_n such that

$$W_n \equiv_{\text{bfo}} A \quad \text{for every } n \in \mathbb{N},$$

and

$$W_n \not\leq_1 W_m \quad \text{whenever } n \neq m.$$

Thus the sets W_n are pairwise incomparable under $\leq_1$, so the corresponding 1-degrees

$$\{\deg_1(W_n) : n \in \mathbb{N}\}$$

form an infinite antichain. Since each $W_n \equiv_{\text{bfo}} A$, all these 1-degrees lie inside the single bounded finite-one degree of A . □

The previous two corollaries isolate the two basic effects of the profile method: constant profiles yield a dense chain, while oscillating profiles force pairwise incomparability. The next corollary shows that these are not separate phenomena attached to different witnesses. By combining the two profile patterns in a single construction, we obtain one bounded finite-one degree that simultaneously contains both a copy of $(\mathbb{Q}, \leq)$ and an infinite antichain of 1-degrees.

Corollary 3.6 (One degree containing both structures). *There exist a nonrecursive set $A \leq_T \emptyset'$, a family $\{B_q\}_{q \in \mathbb{Q} \cap (1,2)}$, and a family $\{W_n\}_{n \in \mathbb{N}}$ such that*

- (i) $B_q \equiv_{\text{bfo}} A$ for every $q \in \mathbb{Q} \cap (1,2)$;
- (ii) $W_n \equiv_{\text{bfo}} A$ for every $n \in \mathbb{N}$;
- (iii) $B_p <_1 B_q$ whenever $p, q \in \mathbb{Q} \cap (1,2)$ and $p < q$;
- (iv) $W_n \not\leq_1 W_m$ whenever $n \neq m$.

Consequently, the bounded finite-one degree of A contains both a copy of $(\mathbb{Q}, \leq)$ and an infinite antichain of 1-degrees.

Proof. Fix an injective computable enumeration

$$q_0, q_1, \dots$$

of $\mathbb{Q} \cap (1,2)$, and fix rationals $1 < \alpha < \beta < 2$. For each $n \in \mathbb{N}$, let

$$Z_n = \{2^n(2t+1) : t \in \mathbb{N}\}.$$

Define a uniformly computable family of profiles $\{u_r\}_{r \in \mathbb{N}}$ by

$$u_{2i}(k) = q_i \quad (i, k \in \mathbb{N}),$$

and

$$u_{2n+1}(k) = \begin{cases} \beta & \text{if } k \in Z_n, \\ \alpha & \text{if } k \notin Z_n \end{cases} \quad (n, k \in \mathbb{N}).$$

Let D consist of the following pairs:

- (a) every pair $(2i, 2j)$ such that $q_i > q_j$;
- (b) every pair $(2n+1, 2m+1)$ such that $n \neq m$.

For pairs of type (a), define

$$\delta_{2i,2j} = q_i - q_j \quad \text{and} \quad s_{2i,2j}(t) = t.$$

For pairs of type (b), define

$$\delta_{2n+1,2m+1} = \beta - \alpha \quad \text{and} \quad s_{2n+1,2m+1}(t) = 2^n(2t+1).$$

No relation is imposed between even and odd indices. We claim that the hypotheses of Theorem 3.2 are satisfied. The family $\{u_r\}_{r \in \mathbb{N}}$ is uniformly computable: for even indices, $u_{2i}(k) = q_i$, and for odd indices, $u_{2n+1}(k)$ is determined by the uniformly computable predicate $k \in Z_n$. The set D is computable. For pairs of type (a), the rational $\delta_{2i,2j} =$

$q_i - q_j > 0$ is uniformly computable, $s_{2i,2j}(t) = t$ is strictly increasing and computable with infinite range, and for every k ,

$$u_{2i}(k) = q_i = q_j + (q_i - q_j) = u_{2j}(k) + \delta_{2i,2j}.$$

For pairs of type (b), the rational $\delta_{2n+1,2m+1} = \beta - \alpha > 0$ is uniformly computable, the function $s_{2n+1,2m+1}(t) = 2^n(2t + 1)$ is strictly increasing and computable with infinite range Z_n , and if $k \in Z_n$, then $k \notin Z_m$, so

$$u_{2n+1}(k) = \beta = \alpha + (\beta - \alpha) = u_{2m+1}(k) + \delta_{2n+1,2m+1}.$$

Thus condition (1) holds for every pair in D . Applying Theorem 3.2 and writing

$$B_{q_i} = C_{2i}, \quad W_n = C_{2n+1},$$

we obtain

$$B_{q_i} \equiv_{\text{bfo}} A \quad \text{and} \quad W_n \equiv_{\text{bfo}} A$$

for all i, n . By items (d) and (e) of Theorem 3.2,

$$B_{q_i} <_1 B_{q_j} \quad \text{whenever } q_i < q_j,$$

and

$$W_n \not\leq_1 W_m \quad \text{whenever } n \neq m.$$

Thus the bounded finite-one degree of A contains both a copy of $(\mathbb{Q}, \leq)$ and an infinite antichain of 1-degrees.

It remains to show that the set A produced by the construction in the proof of Theorem 3.2 is nonrecursive. Fix an index e_{id} for the identity function. For every pair $i, j \in \mathbb{N}$ with $q_i > q_j$, the requirement $R_{e_{\text{id}}, 2i, 2j}$ occurs in that construction. Its vacuous cases are impossible, since the identity is total and injective on every finite set. Hence the construction reaches the diagonalization case, and for fresh blocks $I_k < I_\ell$ it assigns

$$A(z) = 1 \text{ for all } z \in I_k, \quad A(z) = 0 \text{ for all } z \in I_\ell.$$

Since there are infinitely many pairs (i, j) with $q_i > q_j$, this happens infinitely often. Therefore A has infinitely many 1-blocks and infinitely many 0-blocks, so A is infinite and coinfinite. For each i and each k , the block $K_{2i,k}$ is nonempty and

$$h_{2i}[K_{2i,k}] = I_k.$$

Since A is constant on I_k , it follows that $B_{q_i}(x) = A(h_{2i}(x))$ is constant on $K_{2i,k}$, with the same value that A takes on I_k . As A has infinitely many 1-blocks and infinitely many 0-blocks, the set B_{q_i} also has infinitely many 1-blocks and infinitely many 0-blocks. Hence B_{q_i} is infinite and coinfinite.

If A were recursive, then every B_{q_i} would be recursive. But any two infinite coinfinite recursive sets are one-one equivalent. This contradicts the fact that

$$B_{q_i} <_1 B_{q_j} \quad \text{whenever } q_i < q_j.$$

Hence A is nonrecursive. □

Remark 3.7. By Remark 3.3, the 1-degree of A is a common lower bound of the explicitly constructed degrees. More precisely, in Corollaries 3.4 and 3.6 one has

$$A <_1 B_q \quad (q \in \mathbb{Q} \cap (1, 2)),$$

while in Corollaries 3.5 and 3.6 one has

$$A <_1 W_n \quad (n \in \mathbb{N}).$$

Indeed, if $B_q \leq_1 A$, choose $p < q$. Then

$$A \leq_1 B_p <_1 B_q \leq_1 A,$$

a contradiction. Similarly, if $W_n \leq_1 A$, then for any $m \neq n$,

$$W_n \leq_1 A \leq_1 W_m,$$

contradicting $W_n \not\leq_1 W_m$. Thus both the copy of $(\mathbb{Q}, \leq)$ and the infinite antichain constructed in Corollary 3.6 lie strictly above the 1-degree of A .

Remark 3.8. Corollary 3.6 concerns the specific bounded finite-one degree produced by the combined family of constant and oscillating profiles. It does not assert that every bounded finite-one degree containing a dense chain of 1-degrees must also contain an infinite antichain.

Remark 3.9. The witness A from Corollary 3.6 is not m -rigid. Indeed, let $I_k = [a_k, a_{k+1})$ be the computable target blocks from the construction, and recall that A is constant on each I_k . Define $f : \mathbb{N} \rightarrow \mathbb{N}$ by

$$f(x) = a_k \quad \text{whenever } x \in I_k.$$

Then f is total computable, and for every $x \in I_k$ one has

$$x \in A \iff a_k \in A \iff f(x) \in A.$$

Thus f is a computable m -autoreduction of A . However, for every $k \geq 2$ and every $x \in I_k \setminus \{a_k\}$, we have $f(x) \neq x$. Hence f is not eventually the identity, so A is not m -rigid.

3.1. Universal order embeddings. The preceding corollaries show that the profile method can realize both dense chains and infinite antichains inside a single bounded finite-one degree. We now turn to its most general consequence. For each element i , we encode its principal lower set

$$\downarrow i = \{n \in \mathbb{N} : n \preceq i\}$$

into a computable block profile. In this way, pointwise domination of profiles mirrors inclusion of lower sets, and hence the original order relation.

The next corollary may be viewed as the universal form of the oscillating-profile construction. The pairwise disjoint computable sets Z_n serve as independent markers for the elements n of the underlying partial order, and the profile u_i records exactly which markers belong to $\downarrow i$.

Corollary 3.10 (Computable partial orders). *Let $\preceq$ be a computable partial order on $\mathbb{N}$. Then there exist a set $A \leq_T \emptyset'$ and sets*

$$B_0, B_1, \dots$$

such that

$$B_i \equiv_{\text{bfo}} A \quad \text{for every } i \in \mathbb{N},$$

and

$$B_i \leq_1 B_j \quad \text{if and only if } i \preceq j.$$

Hence the function

$$i \mapsto \deg_1(B_i)$$

is an order-embedding of $(\mathbb{N}, \preceq)$ into the partially ordered set of 1-degrees contained in the bounded finite-one degree of A .

Proof. Fix rationals $1 < \alpha < \beta < 2$, and for each $n \in \mathbb{N}$ let

$$Z_n = \{2^n(2t+1) : t \in \mathbb{N}\}.$$

Then the sets Z_n are pairwise disjoint, infinite, and uniformly computable.

For each $i, k \in \mathbb{N}$, define

$$u_i(k) = \begin{cases} \beta & \text{if } k \in Z_n \text{ for some } n \preceq i, \\ \alpha & \text{otherwise.} \end{cases}$$

Since $\preceq$ is computable and the sets Z_n form a uniformly computable pairwise disjoint cover of $\mathbb{N} \setminus \{0\}$, the family $\{u_i\}_{i \in \mathbb{N}}$ is uniformly computable.

Let

$$D = \{(i, j) \in \mathbb{N}^2 : i \not\preceq j\}.$$

Since $\preceq$ is computable, so is D . For each $(i, j) \in D$, define

$$\delta_{i,j} = \beta - \alpha \quad \text{and} \quad s_{i,j}(t) = 2^i(2t + 1).$$

Then $s_{i,j}$ is strictly increasing, computable, and has infinite range

$$S_{i,j} = Z_i.$$

Now fix $(i, j) \in D$, and let $k \in S_{i,j} = Z_i$. Since $\preceq$ is reflexive, we have $i \preceq i$, so

$$u_i(k) = \beta.$$

If $u_j(k) = \beta$, then $k \in Z_n$ for some $n \preceq j$. Since also $k \in Z_i$ and the sets Z_n are pairwise disjoint, it would follow that $n = i$, hence $i \preceq j$, a contradiction. Therefore $u_j(k) = \alpha$. It follows that

$$u_i(k) = \beta = \alpha + (\beta - \alpha) = u_j(k) + \delta_{i,j},$$

so condition (1) holds.

Apply Theorem 3.2 and write

$$B_i = C_i \quad (i \in \mathbb{N}).$$

Then

$$B_i \equiv_{\text{bfo}} A \quad \text{for every } i \in \mathbb{N}.$$

If $i \preceq j$, then for every $k \in \mathbb{N}$ we have $u_i(k) \leq u_j(k)$: indeed, if $u_i(k) = \alpha$ there is nothing to prove, while if $u_i(k) = \beta$, then $k \in Z_n$ for some $n \preceq i$, and by transitivity $n \preceq j$, so $u_j(k) = \beta$. Hence, by item (d) of Theorem 3.2,

$$B_i \leq_1 B_j.$$

Conversely, if $i \not\preceq j$, then $(i, j) \in D$, and item (e) of Theorem 3.2 gives

$$B_i \not\leq_1 B_j.$$

Therefore

$$B_i \leq_1 B_j \quad \text{if and only if} \quad i \preceq j.$$

Since each $B_i \equiv_{\text{bfo}} A$, the corresponding 1-degrees all lie inside the bounded finite-one degree of A . Therefore

$$i \longmapsto \deg_1(B_i)$$

is an order-embedding of $(\mathbb{N}, \preceq)$ into the partially ordered set of 1-degrees contained in the bounded finite-one degree of A . □

The previous corollary gives effective embeddings of computable partial orders into the 1-degrees inside a single bounded finite-one degree. By combining that result with Mostowski's theorem on the existence of a computable universal partial order, we immediately obtain a noneffective universality statement for all countable partial orders. The only additional point is to ensure that the witnessing bounded finite-one degree can still be chosen nonrecursive.

Corollary 3.11 (Universal countable partial orders). *There exists a nonrecursive set $A \leq_T \emptyset'$ such that every countable partial order embeds into the partially ordered set of 1-degrees contained in the bounded finite-one degree of A .*

Proof. Let $\preceq_u$ be a computable universal partial order on $\mathbb{N}$, that is, a computable partial order into which every countable partial order embeds. The existence of such an order is due to Mostowski [6]; see also [13, p. 11] and [8, Exercise V.2.8, p. 461]. Apply Corollary 3.10 to $\preceq_u$. Then there exist a set $A \leq_T \emptyset'$ and sets

$$B_0, B_1, \dots$$

such that

$$B_i \equiv_{\text{bfo}} A \quad \text{for every } i \in \mathbb{N},$$

and

$$B_i \leq_1 B_j \quad \text{if and only if} \quad i \preceq_u j.$$

Every countable partial order thus embeds into

$$\{\deg_1(B_i) : i \in \mathbb{N}\},$$

and hence into the partially ordered set of 1-degrees contained in the bounded finite-one degree of A .

It remains to show that A may be taken nonrecursive. In the construction from the proof of Theorem 3.2, fix an index e_{id} for the identity function. Since $\preceq_u$ is universal, it contains an infinite antichain, so the set

$$D = \{(i, j) \in \mathbb{N}^2 : i \not\preceq_u j\}$$

is infinite. For each $(i, j) \in D$, the requirement

$$R_{e_{\text{id}}, i, j}$$

reaches the diagonalization case, since the identity function is total and injective on every finite set. Hence infinitely many stages assign one fresh block the value 1 and a later fresh block the value 0. Therefore A has infinitely many 1-blocks and infinitely many 0-blocks, so A is infinite and coinfinite.

For each i and each k , the block $K_{i,k}$ is nonempty and

$$h_i[K_{i,k}] = I_k.$$

Since A is constant on I_k , the set $B_i = A \circ h_i$ is constant on $K_{i,k}$, with the same value that A takes on I_k . As A has infinitely many 1-blocks and infinitely many 0-blocks, it follows that each B_i is infinite and coinfinite.

Since $\preceq_u$ is universal, it contains a strict 2-chain, so there exist $p, q \in \mathbb{N}$ such that

$$p \preceq_u q \quad \text{and} \quad q \not\preceq_u p.$$

Hence

$$B_p <_1 B_q.$$

If A were recursive, then every B_i would be recursive. But any two infinite coinfinite recursive sets are one-one equivalent. This contradicts $B_p <_1 B_q$. Therefore A is nonrecursive. □

4. CONCLUDING REMARKS

The results of this paper show that bounded finite-one degrees can support a much richer internal order structure than previously known explicit constructions suggested. In particular, our profile method yields a nonrecursive set $A \leq_T \emptyset'$ whose bounded finite-one degree contains a copy of the dense linear order $(\mathbb{Q}, <)$, an infinite antichain of 1-degrees, and, more generally, an embedded copy of every countable partial order. Thus a single bounded finite-one degree may simultaneously exhibit dense, incomparable, and universal behaviour at the level of 1-degrees.

This picture should be viewed against two complementary background results. On the one hand, Richter, Stephan, and Zhang [13] constructed a nonrecursive bounded finite-one degree whose internal 1-degrees form exactly a strictly ascending chain of order type ω , and asked whether there are bounded finite-one degrees consisting exactly of a dense linearly ordered set of 1-degrees. On the other hand, previous work on m -rigidity [3] shows that, for a measure-1 and comeager class of sets, bounded finite-one degrees already contain infinite antichains of 1-degrees and therefore cannot be linearly ordered. The present construction complements that typical negative picture from the opposite direction: it provides an explicit non- m -rigid witness, necessarily lying in the null, meager complement of the typical rigid regime, whose bounded finite-one degree is far from rigid and already contains a copy of $(\mathbb{Q}, \leq)$.

At the same time, our results do not settle the exact realization problem posed by Richter, Stephan, and Zhang. The bounded finite-one degree constructed here does not consist *exactly* of a dense linear order of 1-degrees, since it also contains incomparable elements. What the profile method shows is that the real difficulty of the problem lies not in producing density itself, but in isolating a degree whose entire internal 1-degree structure is precisely a dense chain. Thus, in the sense of embedded suborders, bounded finite-one degrees can already realize every possible linear order. What remains open, in the sense emphasized by Richter, Stephan, and Zhang in their further remarks, is whether some bounded finite-one degree can have as its entire internal 1-degree structure a linear order other than the one-element order or ω . More broadly, the Profile Theorem isolates a flexible block-geometric mechanism for controlling one-one comparison patterns inside a single bounded finite-one degree. The coexistence of constant and oscillating profiles, as well as the passage from concrete examples to embeddings of arbitrary countable partial orders, shows that, at the level of embedded suborders, the internal order structure of bounded finite-one degrees is considerably richer than had previously been exhibited by explicit constructions.

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